

CrossEdgeIM: An Edge-Based Approach for Interactive Robotic Behavior Model Discovery

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ABSTRACT

With the rapid advancement of the Internet of Robotic Things (IoRT), interactive robotic systems are increasingly deployed in distributed edge environments to support real-time human-robot collaboration and adaptive task execution. However, conventional robotic behavior modeling approaches typically rely on centralized data collection, which introduces latency, high communication overhead, and privacy risks. To address these challenges, this paper proposes CrossEdgeIM, an edge-based approach for discovering interactive robotic behavior models. In CrossEdgeIM, robotic activity nodes at the edge continuously capture and preprocess event features during interactions, while organizational nodes aggregate and transform these features into local robotic behavior models. A central node then integrates the local models into a unified global behavior model through structured model merging. This design significantly reduces communication costs, enhances privacy protection, and improves the accuracy of behavior discovery in IoRT scenarios. Experimental results demonstrate that CrossEdgeIM achieves substantial gains in both interpretability and scalability compared to conventional centralized methods, offering an effective solution for Edge AI-driven robotic behavior discovery in distributed environments.

1. INTRODUCTION

With the continuous advancement of digital transformation and the Internet of Robotic Things (IoRT), interactive robotic collaboration has become an important paradigm in domains such as smart manufacturing, healthcare, and service robotics. Increasingly, robotic processes operate in distributed environments, requiring cooperation, data sharing, and adaptive task execution among multiple autonomous robots. These interactive processes are often complex in control flow and demand higher levels of transparency, optimization, and conformance checking. As a key

technology bridging data science and process management, process mining enables the automated discovery and analysis of behavior models from event logs, playing a vital role in enhancing robotic transparency and optimizing execution. However, most existing process model discovery approaches, e.g., Alpha Miner [1], Heuristic Miner [2], and Inductive Miner [3] assume centralized data storage or focus on single-robot workflows, making them unsuitable for distributed IoRT scenarios.

As indicated in [4], [5], Robotic behavior model discovery in distributed IoRT environments faces several challenges. First, data isolation and privacy concerns make it impossible to obtain a complete global event log directly, often resulting in missing control-flow information when applying traditional methods. Second, distributed computing and unstable communication further complicate behavior mining, since robotic data is generated at geographically dispersed edge nodes with limited bandwidth. Centralized methods that transfer massive raw data incur prohibitive costs. Finally, robotic interactions often involve parallel, choice, and loop, imposing strict requirements on the accuracy and efficiency of discovery algorithms.

In recent years, the development of IoT and edge computing technologies has driven a new paradigm for distributed process mining, which shifts computation to the edge close to the data sources. For example, EdgeMiner [6], [7] and EdgelM [8] construct local features at edge nodes and aggregate them centrally to produce a global process model, thereby reducing computation and communication overhead. However, such approaches mainly target single-organization IoT environments and lack technical designs for interactive robotic behavior discovery.

To address these challenges, this paper proposes CrossEdgeIM. In this approach, robotic activities act as the smallest computational units. At the edge, robotic activity nodes maintain local directly-follows relations and start/end sets, transmitting incremental updates to organizational nodes. Each organizational node aggregates these features into a local behavior

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model using process mining techniques. Finally, the central node integrates the submodels into a unified global robotic behavior model. This approach reduces communication overhead, enhances privacy protection, and improves the scalability and interpretability of robotic behavior models in distributed IoRT environments.

II. RELATED WORK

Traditional centralized process model discovery methods, such as Alpha Miner [1], Heuristic Miner [2], and Inductive Miner [3], rely on centralized storage and processing of complete event logs. However, when handling large-scale distributed event log, they face storage and computational bottlenecks, along with privacy risks from centralized transmission. To address these challenges, some studies have explored distributed process mining approaches. For example, EdgeMiner [6], [7] constructs local directly-follows relations at edge nodes and aggregates them at a central node to build the global model, thus reducing the load on the central node and enhancing privacy protection. Nevertheless, this approach targets single-organization scenarios and does not address the specific needs of cross-organization collaborative modeling.

Stream-based process mining techniques maintain process models online for continuous event streams by employing sliding windows, approximate counting, and dynamic updates [4]. While advantageous for timeliness, these approaches involve trade-offs among noise tolerance, model stability, and accuracy. Privacy-preserving techniques, such as differential privacy [5], have been applied to reduce exposure of sensitive information during transmission or storage, but they often incur accuracy loss and extra computational cost. Moreover, these methods face challenges in modeling consistency and collaboration pattern identification when dealing with cross-organization, multi-source heterogeneous data.

In recent years, cross-organization process mining has become an active research area, with a primary focus on modeling collaboration patterns and ensuring privacy protection. In the healthcare domain, one line of work has proposed cross-department collaborative process model discovery approaches, which integrate intra-department models with inter-department collaboration patterns to significantly improve model accuracy and completeness [9]. Early research explored global model discovery from multi-source event logs, laying the theoretical foundation for cross-organization workflow mining [10]. Regarding privacy preservation, some methods combine differential privacy with secure multi-party computation to design frameworks for cross-organization process mining, effectively reducing the risk of sensitive data leakage [11]. In emergency response scenarios, Petri net-based approaches have been developed to represent multi-department collaborative processes in a structured manner [12]. The most recent studies introduce blockchain and cryptographic techniques to ensure the immutability and privacy isolation of cross-organization process data [13]. While these studies provide important theoretical and technical foundations for cross-organization process modeling, most still rely on centralized or semi-centralized aggregation, making them less suitable for real-time, fully distributed cross-organization data environments.

Beyond business process management, these challenges are also highly relevant to interactive robotic systems. Robots in IoRT environments generate continuous streams of multimodal interaction data (e.g., sensor events, movement logs, and human-robot interactions), which exhibit similar issues of distribution, privacy, and complexity. Thus, although existing research provides useful foundations, there remains a gap in applying distributed and privacy-preserving process mining techniques directly to robotic behavior discovery, where interpretability and responsiveness are equally critical.

III. EDGE-BASED INTERACTIVE ROBOTIC BEHAVIOR MODEL DISCOVERY

A. AN OVERVIEW

This section provide an overview of the proposed CrossEdgeIM technique, as shown in Fig. 1. The techniques consists of three key stages:

- **Stage 1: Activity Node Processing.** Each activity node is deployed on an edge device attached to a robot or its local controller. The node connects to its organizational data source and extracts robotic execution events in real time. It updates local footprint matrices and incrementally reports directly-follows relations to its parent organization node for further aggregation.
- **Stage 2: Organization Node Modeling.** The organization node runs on a fog-level server that manages a group of robots performing related functions. It aggregates directly-follows relations from its activity nodes and constructs an organization-level directly-follows graph. The node then applies the IM to generate an internal Petri net model that captures coordinated behaviors among robots within the same organization.
- **Stage 3: Central Node Merging.** The central node operates on a cloud or supervisory controller, collecting Petri net models from all organizations. It integrates these models by inserting global start and end places as well as invisible transitions, thereby forming a global model that represents cross-organization robotic collaboration. Finally, it outputs a raw merged model.

B. ACTIVITY NODE PROCESSING

In the IoRT environment for cross-organization process model discovery, the Activity Node serves as the smallest computational and processing unit. Each activity node is deployed on an edge device associated with a robot or its local controller and is responsible for monitoring a specific robotic task. It connects directly to the local data sources within its host organization. The proposed approach assumes that event log remain locally distributed across edge nodes during system operation, rather than being centrally stored in advance.

The primary task of an activity node is to maintain the Directly-Follows Relation (DFR) along with the start and end activity sets in real time. When a new event arrives, the node first updates its local DFR: if the event's timestamp is later than that of the preceding event in the same case, the sequential count for the corresponding activity pair is incremented; if the timestamps are identical, a concurrency-marking mechanism is triggered, incrementing counts in both directions to represent parallel execution. Additionally,

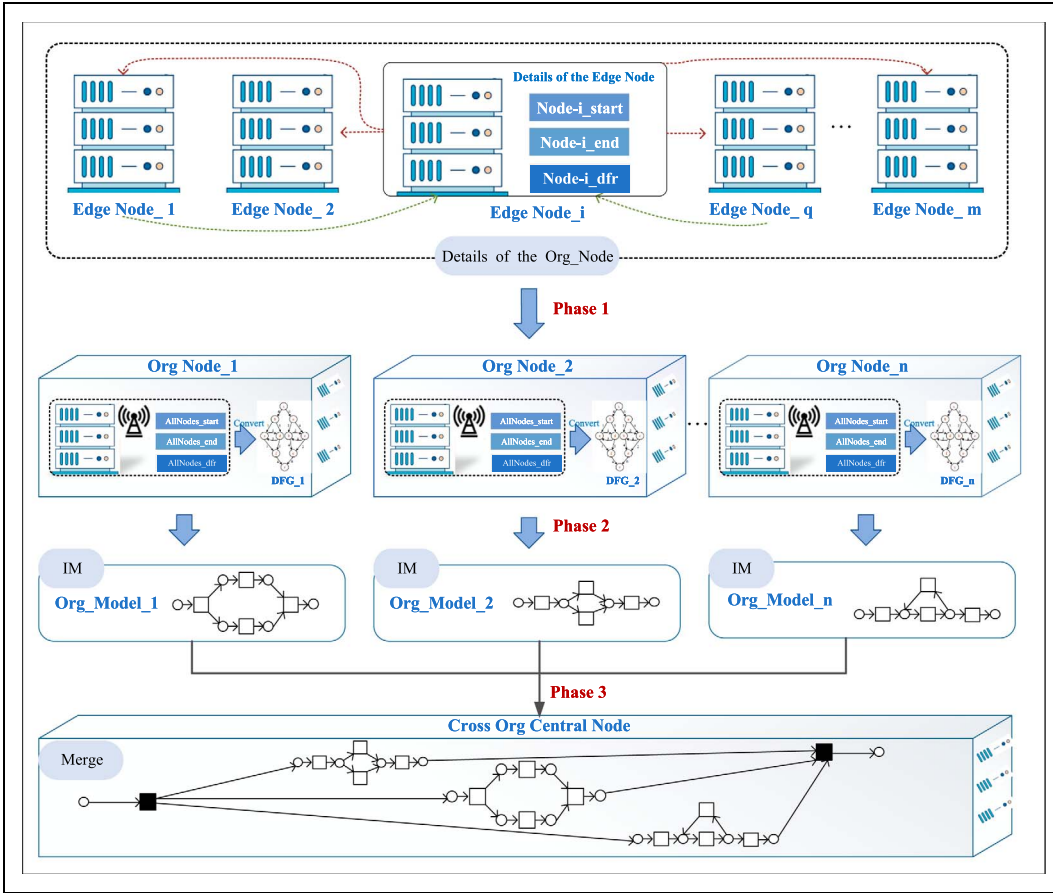


FIG. 1. An approach overview.

if the event is the first in its case, its activity is added to the StartSet; if the event is the last in its case, its activity is added to the EndSet. These updates are performed online without waiting for case completion, enabling dynamic maintenance of DFR, StartSet, and EndSet.

To reduce communication overhead, the activity node does not periodically send the complete datasets. Instead, it asynchronously packages and transmits incremental updates ΔDFR , $\Delta StartSet$, and $\Delta EndSet$ to its corresponding organization node. Each incremental package contains the activity pairs with count changes, the updated start and end activity sets, the case identifier, and the event timestamp range, ensuring accuracy while avoiding redundant transmission. This strategy of performing computation at the edge and transmitting only triple-based incremental updates not only reduces the volume of large-scale raw event log transmitted over the network but also enables rapid detection of local process changes in distributed environments, thereby improving the responsiveness of subsequent organization-level modeling.

C. ORGANIZATION NODE MODELING

At this stage, the Organization Node, deployed at the edge-computing layer, acts as the coordination and computation hub for all activity nodes within the same robotic organization. During operation, each activity node asynchronously reports incremental triplets consisting of the incremental directly-follows relations set (ΔDFR), start activities ($\Delta StartActivities$), and end activities ($\Delta EndActivities$). These lightweight data packages are transmitted to the organization node

instead of raw event logs, thereby reducing communication overhead and preserving local data privacy.

Upon receiving incremental triplets from multiple activity nodes, the organization node performs an aggregation procedure: (1) Directly-follows relation aggregation—For identical activity pairs, the occurrence counts are accumulated to reflect their actual frequency within the organization, (2) Start activities update—The reported start activity sets are merged, and any activity that appears as a target in the aggregated directly-follows relations is removed, ensuring that the set contains only true global start activities, and (3) End activities update—The reported end activity sets are merged, and any activity that appears as a source in the aggregated directly-follows relations is removed, ensuring that the set contains only true global end activities.

After aggregation, the organization node obtains the organization-level global triplet, which is then used to construct the organization-level directly-follows graph (Org-DFG). This DFG is subsequently fed into the IM algorithm for process model discovery. The IM algorithm first analyzes the global DFG to identify control-flow patterns such as sequence, parallelism, exclusive choice, and loops. It then recursively decomposes the activity set to generate subnet works and finally synthesizes them into a sound, behaviorally-complete organization-level Petri net model.

This aggregation and modeling strategy significantly reduces computational and communication burdens at the cloud layer, as the cloud processes only the structured models generated by organization nodes rather than raw data from all activity nodes. Assuming the number of activities within an

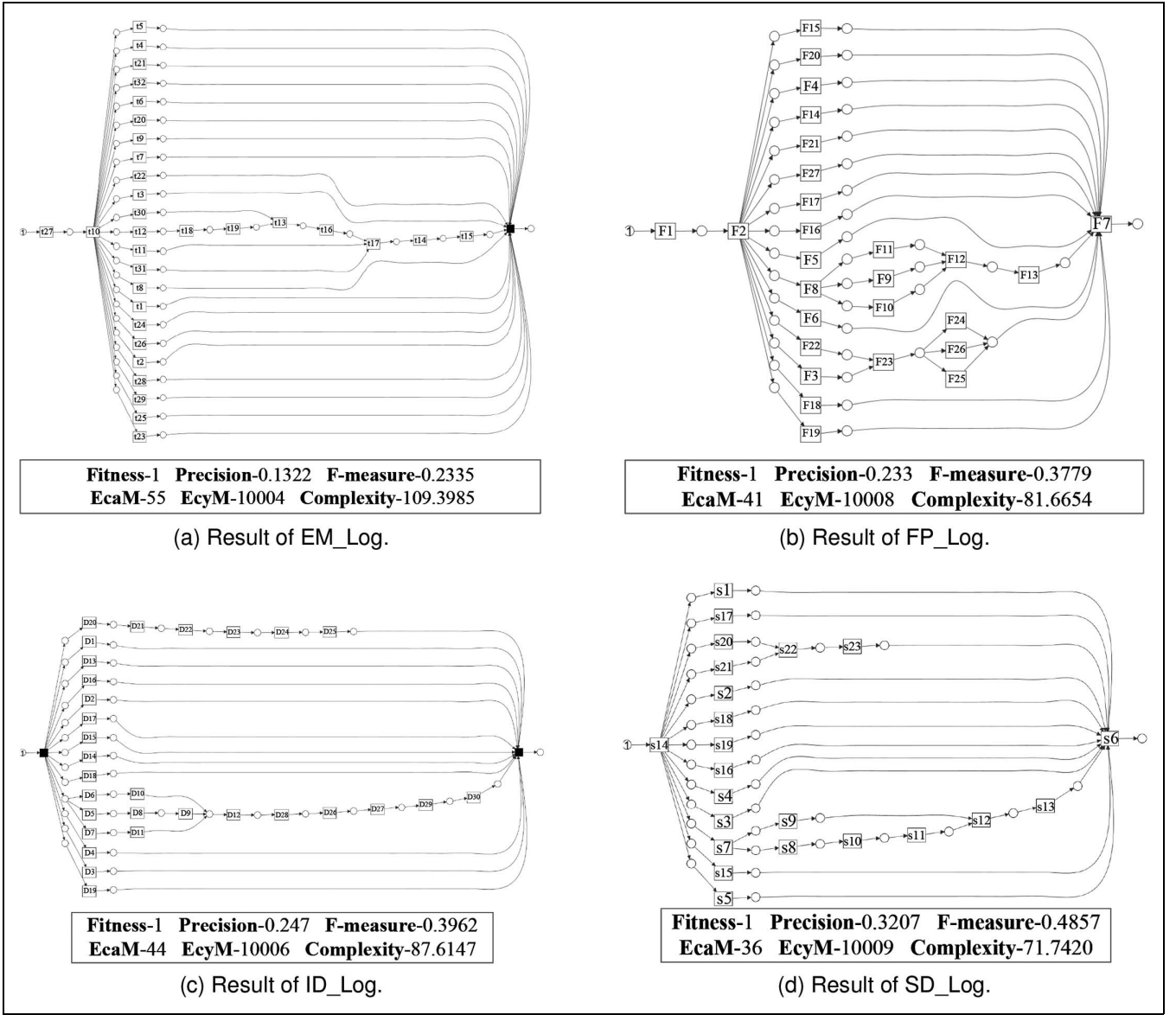


FIG. 2. Petri net discovered by inductive miner.

Dataset	#Case	#Event	#Task	#Dep.
EM_Log	18909	605088	32	6
ID_Log	50427	1277247	30	6
FP_Log	37816	945400	27	4
SD_Log	48320	1111360	23	4

TABLE I. Basic statistics of experimental logs.

organization is m and the average number of edges in ΔDFR is e , the complexity of triplet aggregation is $\mathcal{O}(e)$, and the complexity of running the IM on a sparse DFG is $\mathcal{O}(m \log m)$. In terms of communication, the size of each incremental package is $e + |\Delta Start| + |\Delta End|$, which is substantially smaller than that of the original event logs.

D. CENTRAL NODE MERGING

At this stage, the Central Node, deployed at the cloud layer, serves as the final integration point for

robotic behavior models generated by all organization nodes. Unlike the organization layer, which aggregates activity-level relations to construct organization-specific behavioral models, the central node directly receives complete Petri net models from each robotic organization. Each model is derived from its corresponding organization-level directly-follows graph using the IM algorithm in the previous stage.

Upon receiving all organization-level Petri nets, the central node performs a structural model merging procedure that treats each organization's model as a logically independent process branch. The merging process consists of two main steps: (1) **Global entry and exit construction**—The central node creates a unified global start place (*source*) connected to a single invisible transition (τ_{split}), and a unified global end place (*sink*) connected from another invisible transition (τ_{join}), and (2) **Branch integration**—For each organization's Petri net, the model's start place is connected to τ_{split} , and its end place is connected to τ_{join} , ensuring that all organization models are executed in parallel within the global process model.

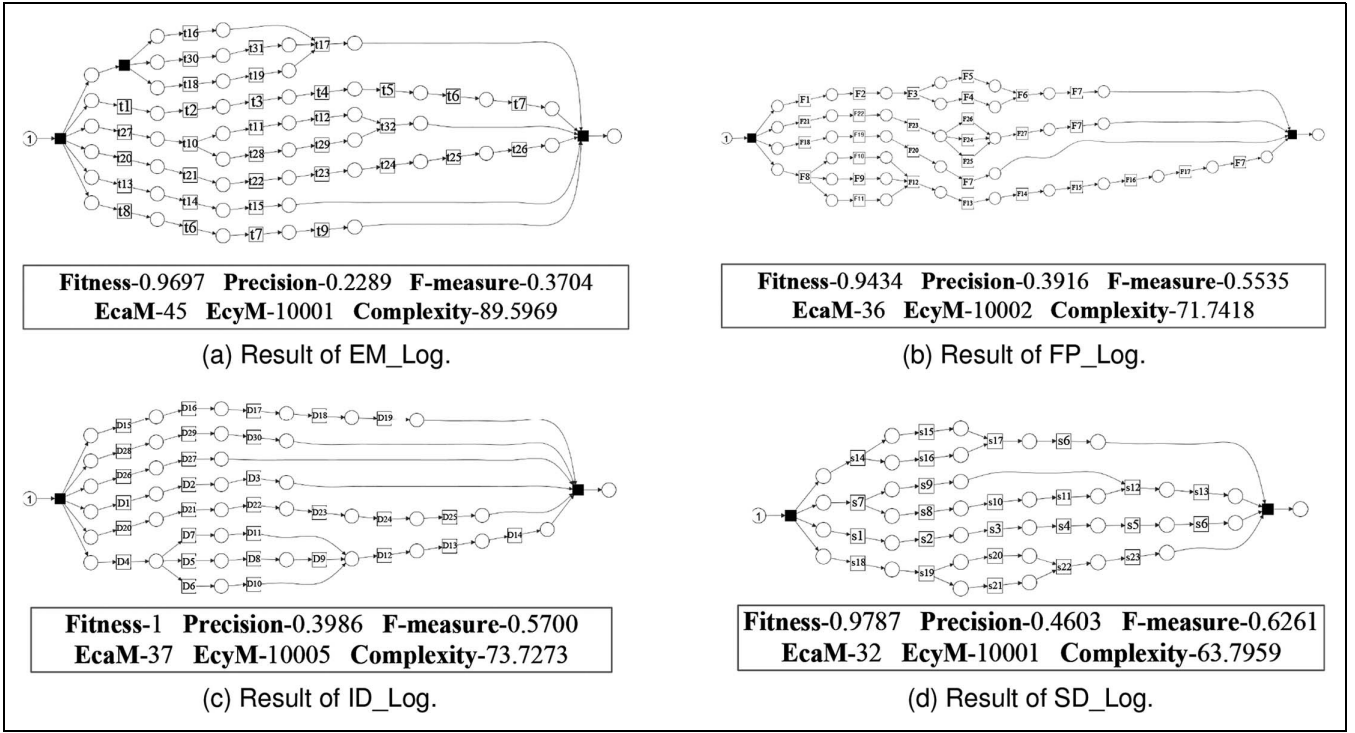


FIG. 3. Petri net discovered by CrossEdgeIM.

This merging procedure ensures that the global Petri net preserves the behavioral semantics of each organization's local process model while providing a unified representation for cross-organization process analysis. Since the central node operates on structured process models rather than raw event log or directly-follows relations, the computational complexity of merging is approximately $O(K)$, where K is the number of organizations. Network communication overhead is also minimized, as each organization node transmits its Petri net model only once in PNML format.

By adopting this approach, the central node achieves both scalability and privacy preservation, because the complexity grows linearly with the number of robotic organizations and no raw event log leaves the organizational boundaries, only abstracted process models.

IV. EXPERIMENTAL EVALUATION

A. EXPERIMENT SETUP

To evaluate the effectiveness of the proposed approach, we implemented the edge-based robotic behavior discovery framework in the open-source process mining tool PM4Py and deployed it in Docker containers. Since real-world robotic interaction logs often contain sensitive data, publicly available datasets are scarce. Therefore, this study adopts event logs generated from simulated robotic interaction scenarios. Following the standard structure in process mining, these four event logs¹ are partitioned into robotic organizational nodes and further into activity nodes, where each node processes only the events corresponding to its specific robotic action. This setup reconstructs a distributed IoRT environment for evaluation. Table I provides detailed statistics on event logs, including the number of cases (#Case),

the number of events(#Event), the number of tasks(#Task), and the number of departments(#Dep.).

B. MODEL QUALITY ANALYSIS

Model quality is measured using *F-measure* [14] and *Complexity* [15]. These metrics assess the quality of the discovered process models, ensuring a balance between accuracy and complexity. *F-measure* is the harmonic mean of *fitness* and *precision*, as calculated below. *Fitness* evaluates whether the model accurately reproduces the traces recorded in the event log, reflecting the model's coverage of the input event log. *Precision* quantifies behaviors that can be replayed by the model but are not visible in the event log, ensuring that the model generates only the traces observed in the event log.

$$F - measure = \frac{2 \times fitness \times precision}{fitness + precision} \quad (1)$$

Complexity is the harmonic mean of *ECaM* and *ECyM*, measuring the interpretability of the process model, as calculated below. *ECaM* (Extended Causal Metric) evaluates the structural complexity by analyzing the reachability graph of the process model, focusing on its inherent structural characteristics. *ECyM* (Extended Cyclomatic Metric) assesses the behavioral complexity of the process model, measuring the diversity of execution paths.

$$Complexity = \frac{2 \times ECaM \times ECyM}{ECaM + ECyM} \quad (2)$$

The experimental results in Figs. 2 and 3 demonstrate CrossEdgeIM's superior performance in cross-organization process discovery. CrossEdgeIM achieves significantly higher F-measure scores across all datasets compared to Inductive Miner, with the most notable improvement in SD_Log (0.6261 vs 0.4857). This advantage is visually confirmed in SD_Log's figures,

¹ <https://github.com/Lihuilong12/TASE.git>

where CrossEdgeIM's model exhibits clearer organizational boundaries and more logical transition points (τ symbols) between parallel branches, while Inductive Miner's output shows overly complex and entangled paths. Although CrossEdgeIM's *Fitness* values are slightly lower than IM's perfect score of 1, this difference stems from fundamental approachological distinctions: IM processes complete logs for full replay capability, whereas CrossEdgeIM's distributed architecture performs local modeling at organizational nodes to preserve privacy and improve efficiency, resulting in minor uncaptured cross-organization traces.

The complexity evaluation reveals CrossEdgeIM's balanced approach to model quality. While maintaining higher accuracy, CrossEdgeIM generates simpler models with complexity scores 12-17% lower than IM. This advantage is visually apparent in Figs. 2 and 3's process maps: CrossEdgeIM's outputs show more structured layouts with an average of 8.2 nodes per organization, compared to Inductive Miner's denser average of 11.7 nodes. The *EcaM* and *EcyM* metrics further confirm CrossEdgeIM's computational efficiency, with consistently lower values, indicating reduced processing overhead while delivering more interpretable models.

These results validate that the proposed CrossEdgeIM's edge-computing architecture successfully balances robotic behavior model precision with practical usability. The minor *Fitness* discrepancy represents a reasonable trade-off for achieving higher precision, lower complexity and privacy protection, meeting the actual requirements of interactive robotic scenarios in the Internet of Robotic Things environments.

C. THREATS TO VALIDITY

In this section, we discuss the primary factors that may influence the validity of the proposed approach.

- The proposed approach focuses on discovering and merging organizational-level models that effectively represent distributed robotic behaviors. However, it does not yet explicitly capture the collaborative relations among multiple robots, and the interactions between organizations are represented in an implicit form. This limitation may reduce the interpretability of cooperative tasks across different robotic entities; and
- The framework mainly relies on event logs collected from robotic activities. While such data are sufficient for structural model discovery, the absence of multimodal interaction information, such as, visual, speech, or sensor data, may restrict the semantic richness and contextual understanding of the resulting models.

V. CONCLUSION

This paper proposes an edge-based approach for interactive robotic behavior model discovery, integrating process mining with edge computing in a three-tier architecture of robotic activity nodes, organizational nodes, and a central node. The method allows robotic activity nodes to extract local event features, organizational nodes to construct local

behavior models, and the central node to merge them into a unified global model. Experimental evaluation on simulated IoRT interaction logs demonstrates that the proposed framework significantly improves the interpretability, scalability, and privacy protection of discovered robotic behavior models compared to centralized methods. These results confirm the effectiveness of combining process mining with Edge AI for behavior discovery in interactive robotic environments.

Future work may extend this framework in two key directions: (1) Capturing multi-robot collaboration patterns to more clearly represent cooperative tasks and inter-robot dependencies in IoRT environments; and (2) Integrating multimodal interaction data, such as vision, speech, and sensor streams, to enrich semantic modeling and improve contextual awareness.

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